NovoTech MACHINE TOOLS Inc.

PRODUCT CATALOGUE 2026

SENFENG LASER · MADE IN CHINA · SOLD & SERVICED IN CANADA BY NOVOTECH

LASER WELDING AND CLADDING MACHINES

SENFENG HANDHELD LASER WELDERS 0.8 – 3 KW · ROBOTIC LASER WELDING CELLS 1.5 – 20 KW · COBOT MIG WELDERS · LASER CLADDING 6 / 12 KW







China
MADE IN CHINA

3
MACHINE FAMILIES

1-year
WARRANTY

Canada
PARTS & TECHNICAL SUPPORT

SENFENG LASER WELDING

CONTENTS

HANDHELD LASER WELDERS

Handheld laser welders, 0.8 – 3 kW — portable air-cooled S series and HWM series — SF3S · SF4S · SF5S · SF1500HWM · SF2000HWM · SF3000HWM — 04

ROBOTIC WELDING

Robotic laser welding cells, 1.5 – 20 kW, and robotic arc welders — SF1500RW – SF20000RW · SF-500RMW · 350CMW · 500CMW — 07

LASER CLADDING

Laser cladding machines, 6 / 12 kW — shafts, rolls, bores and on-site repair — SFR6063 · SFHS3050 · SFHS6050 · SFMR02 — 10

Specifications subject to change without notice. Imperial values are converted from the manufacturer's metric data and rounded: thin sheet in gauge (ga), plate and capacities in common fractions (1/2, 1/4, 1/8, 1/16); metric values in brackets are the manufacturer's nominal figures. Model numbers are manufacturer designations. Photos may show optional equipment.

ABOUT NOVOTECH

YOUR CANADIAN SOURCE FOR SENFENG LASER WELDING AND CLADDING MACHINES

Sales, installation, parts and technical support across Canada.

NovoTech Machine Tools Inc. supplies metalworking and fabrication machinery to job shops, fabricators, maintenance departments and manufacturers across Canada from our base in Mississauga, Ontario, Canada. Every machine in this catalogue is sold and supported locally by NovoTech — with spare parts and technical support readily available in Canada.

INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in China by Senfeng Laser — CE, FDA and ETL certified and CSA approved
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

SENFENG LASER · JINAN, CHINA

FIVE TIMES FASTER THAN TIG, WITH ALMOST NO HEAT IN THE PART

A fiber laser welder joins stainless, aluminum, galvanized and mild steel sheet with a narrow, deep seam at five times the speed of TIG, with almost no distortion and no grinding afterwards; an operator learns the handheld gun in a day. The portable S series runs 0.8 to 1.5 kW air-cooled from a 220 V outlet; the HWM welds to 9/32″ with 1.5 to 3 kW, a wire feeder and a 4-in-1 head; the RW cells put 1.5 to 20 kW on a 6-axis robot with positioners and tracks; the CMW cobot and the teaching-free RMW bring MIG robots to shops that have never programmed one. The laser cladding machines rebuild shafts, rolls, cylinders and bores with a metallurgically bonded wear layer — in a robot cell, on a lathe bed, or on site.

Senfeng Laser has built fiber laser machines in Jinan, Shandong, since 2004: a 1.8 million ft^2 campus, its own laser sources, cutting heads and welding guns, around 15,000 machines a year, 20 overseas branches and Senfeng Laser USA with a warehouse and service team. CE, FDA and ETL certified and CSA approved. NovoTech sells, installs and supports the machines in Canada from Mississauga, ON, Canada — layout for your building, installation, operator training, a CSA-approved panel, a 1-year warranty and parts and technical support in Canada.

SF3S · SF4S · SF5S · SF1500HWM · SF2000HWM · SF3000HWM

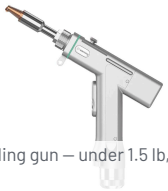
HANDHELD LASER WELDERS, 0.8 – 3 KW — PORTABLE AIR-COOLED S SERIES AND HWM SERIES



MODEL S · HWM · SENFENG HWM SERIES HANDHELD FIBER LASER WELDER — 1.5, 2 OR 3 KW, WELDS TO 9/32″



Senfeng S series portable air-cooled laser welder with wire feeder — 0.8 – 1.5 kW



Handheld laser welding gun — under 1.5 lb, wire feeder and air-cooled head

A handheld laser welder replaces TIG on stainless, aluminum, galvanized and carbon steel sheet: a 1.5 lb gun, a seam five times faster than TIG, almost no heat distortion and no grinding afterwards — an operator learns it in a day. The S series is the portable one: 0.8 to 1.5 kW, air-cooled, 99 to 121 lb on wheels, plugged into a 220 V outlet. The HWM series brings 1.5 to 3 kW with a water chiller and a wire feeder for welds to 9/32″, and a 4-in-1 head that welds, cleans, cuts and seam-welds. CE face shield, protective clothing and an optional welding booth come with it.

STANDARD FEATURES

- ✓ S series: 0.8 / 1.2 / 1.5 kW air-cooled, 99 – 121 lb; HWM: 1.5 / 2 / 3 kW, welds to 9/32″, wire feeder
- ✓ 1.5 lb (0.7 kg) welding gun with wire feed; 16 – 33 ft cable
- ✓ Self-developed laser source and 4-in-1 welding head (weld, clean, cut, seam)
- ✓ Safety: CE face shield, laser-resistant clothing and gloves supplied; optional booth with protective glass
- ✓ CE / FDA certified and CSA approved; S series 220 V single-phase, HWM 3-phase

OPTIONAL EQUIPMENT

- + Laser welding booth
- + Robot + handheld combination (robot CRP-RH18-20 with the same source)
- + Cleaning head; extra wire feeder

S SERIES PORTABLE AIR-COOLED LASER WELDERS — TECHNICAL SPECIFICATIONS

Model	SF3S	SF4S	SF5S
Laser power	0.8 kW	1.2 kW	1.5 kW
Cooling	air-cooled — no chiller	air-cooled — no chiller	air-cooled — no chiller
Wavelength	1,080 ± 10 nm	1,080 ± 10 nm	1,080 ± 10 nm
Filler wire	0.03 – 0.06″ (0.8 – 1.6 mm)	0.03 – 0.06″ (0.8 – 1.6 mm)	0.03 – 0.06″ (0.8 – 1.6 mm)
Welding gun	under 1.5 lb (0.7 kg)	under 1.5 lb (0.7 kg)	under 1.5 lb (0.7 kg)
Gun cable	16.4 ft (5 m)	23 ft (7 m)	23 ft (7 m)
Machine weight	99 lb (45 kg)	110 lb (50 kg)	121 lb (55 kg)
Dimensions L × W × H	26.4 × 13.2 × 23.2″ (670 × 335 × 590 mm)	26.4 × 13.2 × 23.2″ (670 × 335 × 590 mm)	26.4 × 13.2 × 23.2″ (670 × 335 × 590 mm)
Power supply	220 V single-phase	220 V single-phase	220 V single-phase

DETAILS, COMPONENTS AND OPTIONS

HANDHELD LASER WELDERS, 0.8 – 3 KW — PORTABLE AIR-COOLED S SERIES AND HWM SERIES



S series air-cooled welding gun

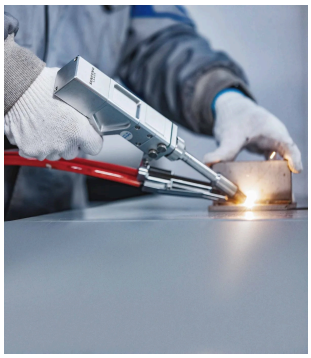


HWM series laser welder with the wire feeder

Handheld laser welding a stainless corner — no filler, no grinding



Handheld laser welding with filler wire



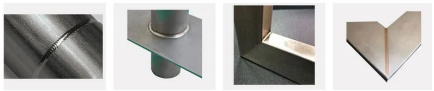
Handheld laser welding a stainless sheet



Operator in laser-resistant clothing, gloves and face shield welding with the handheld gun



Senfeng's laser welding centre — handheld and robotic welding stations



Laser-welded seams on stainless, aluminum and carbon steel

TECHNICAL SPECIFICATIONS

HANDHELD LASER WELDERS, 0.8 – 3 KW — PORTABLE AIR-COOLED S SERIES AND HWM SERIES

HWM SERIES HANDHELD LASER WELDERS

3 models · 1.5 – 3 kW · welds to 9/32″

Model	SF1500HWM	SF2000HWM	SF3000HWM
Laser power	1.5 kW	2 kW	3 kW
Welding thickness	to 5/32″ (4 mm)	to 3/16″ (5 mm)	to 9/32″ (7 mm)
Wavelength	1,080 ± 10 nm	1,080 ± 10 nm	1,080 ± 10 nm
Filler wire	0.03 – 0.06″ (0.8 – 1.6 mm)	0.03 – 0.06″ (0.8 – 1.6 mm)	0.03 – 0.08″ (0.8 – 2 mm)
Welding gun	under 1.5 lb (0.7 kg) · 4-in-1 head: weld, clean, cut, seam	under 1.5 lb (0.7 kg) · 4-in-1 head: weld, clean, cut, seam	under 1.5 lb (0.7 kg) · 4-in-1 head: weld, clean, cut, seam
Gun cable	32.8 ft (10 m)	32.8 ft (10 m)	32.8 ft (10 m)
Machine weight	485 lb (220 kg)	485 lb (220 kg)	584 lb (265 kg)
Dimensions L × W × H	37.6 × 28.2 × 42.5″ (954 × 715 × 1,080 mm)	37.6 × 28.2 × 42.5″ (954 × 715 × 1,080 mm)	47.6 × 28.2 × 42.5″ (1,210 × 715 × 1,080 mm)
Power supply	208 / 240 / 480 / 600 V 3-phase 60 Hz (transformer supplied); single-phase not available	208 / 240 / 480 / 600 V 3-phase 60 Hz (transformer supplied); single-phase not available	208 / 240 / 480 / 600 V 3-phase 60 Hz (transformer supplied); single-phase not available

Welding thickness is for a square butt weld in mild steel; stainless and aluminum run about 70 % of it.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

INCLUDED WITH EVERY NOVOTECH MACHINE

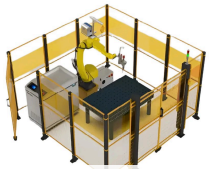
- ✓ Made in China by Senfeng Laser — CE, FDA and ETL certified and CSA approved
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

SF1500RW – SF20000RW · SF-500RMW · 350CMW · 500CMW

ROBOTIC LASER WELDING CELLS, 1.5 – 20 KW, AND ROBOTIC ARC WELDERS



MODEL RW · RMW · CMW · SENFENG RW SERIES ROBOTIC FIBER LASER WELDING CELL — 6-AXIS ROBOT WITH A 1.5 – 20 KW SOURCE



RW series robot laser welder with positioner



Robotic laser welding cell with the horizontal positioner for tanks and tubes

The RW series puts a fiber laser on a 6-axis robot with 72″ reach: 1.5 to 20 kW for sheet to 9/16″, with horizontal or dual-axis positioners for tanks and tubes and 7 to 9-axis cells on a linear track. For MIG/MAG the RMW is a teaching-free robot that scans the part and programs the weld itself, and the CMW is a collaborative robot on a mobile cart with a 59 × 59″ 3D fixture table — 350 or 500 A, programmed by hand-guiding, moved to the job on wheels.

STANDARD FEATURES

- ✓ RW: 6-axis robot, 1.5 – 20 kW, welds to 9/16″ (14 mm); positioners and linear tracks
- ✓ RMW: teaching-free 500 A MIG/MAG; CMW: cobot 350 / 500 A on a cart with 3D fixture table
- ✓ Safety fencing, light curtains and fume extraction interfaces
- ✓ CE, FDA and ETL certified and CSA approved; 208 / 240 / 480 / 600 V 3-phase 60 Hz (transformer supplied); single-phase not available

OPTIONAL EQUIPMENT

- + Horizontal or dual-axis positioner
- + 7 – 9 axis cells with linear track
- + Second welding station; fume extraction

RW SERIES ROBOTIC LASER WELDING CELLS — TECHNICAL SPECIFICATIONS

Model	SF1500RW	SF2000RW	SF3000RW	SF6000RW	SF12000RW	SF20000RW
Laser power	1.5 kW	2 kW	3 kW	6 kW	12 kW	20 kW
Welding thickness	to 0.12″ (3 mm)	to 0.16″ (4 mm)	to 0.24″ (6 mm)	to 0.28″ (7 mm)	to 0.39″ (10 mm)	to 0.55″ (14 mm)
Welding head	to 2.2 kg · wire feeder	to 2.2 kg · wire feeder	to 2.2 kg · wire feeder	to 8 kg · wire feeder	to 8 kg · wire feeder	to 8 kg · wire feeder
Robot	6 axes · 72″ (1,830 mm) reach · 25 kg payload	6 axes · 72″ (1,830 mm) reach · 25 kg payload	6 axes · 72″ (1,830 mm) reach · 25 kg payload	6 axes · 72″ (1,830 mm) reach · 25 kg payload	6 axes · 72″ (1,830 mm) reach · 25 kg payload	6 axes · 72″ (1,830 mm) reach · 25 kg payload
Fiber length	32.8 ft (10 m)	32.8 ft (10 m)	65.6 ft (20 m)	65.6 ft (20 m)	65.6 ft (20 m)	65.6 ft (20 m)
Options	horizontal or dual-axis positioner, 7 – 9 axis cell with linear track, safety fencing	horizontal or dual-axis positioner, 7 – 9 axis cell with linear track, safety fencing	horizontal or dual-axis positioner, 7 – 9 axis cell with linear track, safety fencing	horizontal or dual-axis positioner, 7 – 9 axis cell with linear track, safety fencing	horizontal or dual-axis positioner, 7 – 9 axis cell with linear track, safety fencing	horizontal or dual-axis positioner, 7 – 9 axis cell with linear track, safety fencing

DETAILS, COMPONENTS AND OPTIONS

ROBOTIC LASER WELDING CELLS, 1.5 – 20 KW, AND ROBOTIC ARC WELDERS



Fully enclosed robotic laser welding workstation



Senfeng RMW series teaching-free robotic arc welding machine — 500 A MIG/MAG



Senfeng CMW series collaborative robotic arc welder on a mobile cart — 350 / 500 A



CMW collaborative welder — the cobot arm and torch

TECHNICAL SPECIFICATIONS

ROBOTIC LASER WELDING CELLS, 1.5 – 20 KW, AND ROBOTIC ARC WELDERS

ROBOTIC ARC WELDING MACHINES

3 models · 350 – 500 A MIG/MAG

Model	SF-500RMW	350CMW	500CMW
Type	teaching-free robotic welder — scans the part and programs itself	collaborative cobot welder on a mobile cart	collaborative cobot welder on a mobile cart
Welding source	500 A MIG/MAG, water-cooled torch	350 A, air-cooled torch	500 A, water-cooled torch
Robot	6 axes · 83″ (2,116 mm) reach · 6 kg	cobot · 53″ (1,350 mm) reach · 3 – 20 kg	cobot · 53″ (1,350 mm) reach · 3 – 20 kg
Repeat accuracy	±0.003″ (±0.08 mm)	±0.001″ (±0.03 mm)	±0.001″ (±0.03 mm)
Welding table	3D fixture table (custom)	59 × 59 × 26″ 3D flexible table	59 × 59 × 26″ 3D flexible table
Total power	—	16.5 kVA	27 kVA

Robot reach and payload as listed; the RW cell is sized to the part at order.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in China by Senfeng Laser — CE, FDA and ETL certified and CSA approved
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

SFR6063 · SFHS3050 · SFHS6050 · SFMR02

LASER CLADDING MACHINES, 6 / 12 KW — SHAFTS, ROLLS, BORES AND ON-SITE REPAIR



MODEL SFR · SFHS · SFMR · SENFENG SFR6063 ROBOT LASER CLADDING MACHINE — 6 / 12 KW, WORKPIECES TO 20 FT AND 15 T



Senfeng SFHS3050 laser cladding machine for shafts to 10 ft



Senfeng SFMR02 portable laser cladding system for on-site repair

Laser cladding lays a metallurgically bonded wear or corrosion layer — up to 25/64″ thick — onto shafts, rolls, hydraulic cylinders, valves and dies with far less heat than overlay welding and far better bonding than spraying. The SFR6063 robot cell handles workpieces to 20 ft and 15 t on its table; the SFHS3050 and 6050 are lathe-type machines for shafts to 10 and 20 ft; the SFMR02 is the portable system that goes to the machine for on-site repair. All with Senfeng's high-speed cladding head, dual-cylinder powder feeder and an inner-bore head as an option.

STANDARD FEATURES

- ✓ 6 or 12 kW; high-speed cladding head; dual-cylinder pneumatic powder feeder, 20 – 300 µm, to 150 g/min
- ✓ KUKA / FANUC robot (SFR, SFMR) or lathe bed with Ø19.7″ three-jaw chuck (SFHS)
- ✓ Coatings to 25/64″ (10 mm) with metallurgical bonding and small heat-affected zone
- ✓ CE, FDA and ETL certified and CSA approved; 208 / 240 / 480 / 600 V 3-phase 60 Hz (transformer supplied); single-phase not available

OPTIONAL EQUIPMENT

- \+ Inner-bore cladding head
- \+ Dual-axis positioner
- \+ Powder recovery and fume extraction

LASER CLADDING MACHINES — TECHNICAL SPECIFICATIONS

Model	SFR6063	SFHS3050	SFHS6050	SFMR02
Type	robot cell with KUKA / FANUC robot	lathe-type machine	lathe-type machine	portable robot system for on-site repair
Laser power	6 / 12 kW	6 / 12 kW	6 / 12 kW	6 kW
Max. workpiece length	236″ (6,000 mm)	118″ (3,000 mm)	236″ (6,000 mm)	robot reach 71″ (1,813 mm)
Max. swing · clamping Ø	Ø59 · Ø25″	Ø31.5 · Ø31.5″	Ø31.5 · Ø31.5″	—
Max. table load	33,000 lb (15 t)	4,400 lb (2 t)	4,400 lb (2 t)	—
Powder feeder	dual-cylinder pneumatic feeder, 20 – 300 µm powder, to 150 g/min	dual-cylinder pneumatic feeder, 20 – 300 µm powder, to 150 g/min	dual-cylinder pneumatic feeder, 20 – 300 µm powder, to 150 g/min	dual-cylinder pneumatic feeder, 20 – 300 µm powder, to 150 g/min
Cladding head	high-speed cladding head; inner-bore head optional	high-speed cladding head; inner-bore head optional	high-speed cladding head; inner-bore head optional	high-speed cladding head; inner-bore head optional

Workpiece sizes are the machine limits; cladding rates depend on the powder and layer thickness.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

DETAILS, COMPONENTS AND OPTIONS

LASER CLADDING MACHINES, 6 / 12 KW – SHAFTS, ROLLS, BORES AND ON-SITE REPAIR



KUKA / FANUC robot of the cladding cell



Dual-axis positioner for cladding



High-speed laser cladding head with powder feed



Inner-bore cladding head

LET'S FIND THE RIGHT MACHINE FOR YOUR SHOP.

Call or visit us for quotes, tooling, demonstrations, installation and service. In-stock machines ship fast from our Mississauga, ON, Canada location.

PHONE

+1 647 526 5510

WEB & EMAIL

NovoTechMachineTools.com
info@NovoTechMachineTools.com

ADDRESS

1365 Mid-Way Blvd, Unit 26
Mississauga, ON L5T 2J5

NT NovoTech MACHINE TOOLS Inc.

Specifications subject to change without notice. Imperial values are converted from manufacturer metric data and rounded: thin sheet in gauge (ga), plate and capacities in common fractions (1/2, 1/4, 1/8, 1/16). Model numbers are manufacturer designations. Photos may show optional equipment. 7-day money-back guarantee on in-stock machines only; financing available.